The Informativeness of Frequency-Report Scoring Rules

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Abstract

In a multinomial frequency-report environment, a subject has latent beliefs p, but the researcher observes only an incentivized count report r. We compare frequency-report scoring rules by how sharply that report identifies the latent beliefs. The inference is recast as a partial-identification problem: each scoring rule maps the report to an identified set of beliefs—those under which the report would have been optimal—from which coordinate and mean bounds follow. The main analytical result is that squared-distance frequency scoring gives a transparent projection rule, a linear identified set, closed-form coordinate bounds, and linear-program bounds for means. Frequency-guessing scoring is the known fixed-prize rule with closed-form bounds, and Manhattan distance has an exact identified set with sharp semi-analytical bounds. No rule is universally most informative. Which one yields the tightest bounds turns on how concentrated the subject's beliefs are: squared-distance scoring when beliefs are concentrated on a few categories, frequency-guessing scoring when they are balanced across them, with a crossover between the regimes that is established analytically for the closed-form rules and quantified by a design comparison. Manhattan distance is rarely the tightest, but its bounds barely move across regimes, making it the robust choice when the belief regime cannot be anticipated.

Keywords: belief elicitation; scoring rules; frequency reports; partial identification; informativeness; multinomial beliefs.

JEL classification: C44; C81; C90; D81; D83.

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1 Introduction

Belief elicitation in experiments often asks subjects to report probabilities directly. Frequency reports ask instead for counts: out of n realizations, how many times will each outcome occur? This format is natural when the object of interest is a distribution of choices, signals, or outcomes in a finite group. It may also be easier for subjects than reporting abstract probabilities under a standard scoring rule. Many probability scoring rules, including the quadratic scoring rule, are well understood theoretically (Savage, 1971; Gneiting and Raftery, 2007; Selten, 1998), but they require subjects to understand how probability reports map into payments. That mapping can be cognitively demanding, especially when subjects reason more naturally in frequencies than in abstract probabilities (Hogarth, 1975; Gigerenzer and Hoffrage, 1995; Schlag et al., 2015). Direct monetary scoring rules also raise the familiar risk-aversion problem (Armantier and Treich, 2013).

The practical problem is that the researcher observes a count report r, not the subject's latent belief vector p. The report is an incentivized message generated by a scoring rule. Thus the inferential question is inverse: given the rule and the observed report, which beliefs are consistent with optimal reporting? The paper follows the chain

$$S \;\to\; R_S(p) \;\to\; P_S(r) \;\to\; \big[\underline{p}_{S,i}(r), \overline{p}_{S,i}(r)\big].$$

Here S is a frequency-report scoring rule, $R_S(p)$ is the set of reports that are optimal for a subject with beliefs p, and

$$P_S(r) = \{p \in \Delta^k : r \in R_S(p)\}$$

is the identified set induced by the observed report. This recasts incentivized belief elicitation as a partial-identification problem (Manski, 2003): the scoring-rule mechanism, rather than a sampling distribution over repeated observations, supplies the identifying restriction. The final object of the chain is the bound $[\underline{p}_{S,i}(r), \overline{p}_{S,i}(r)]$ it yields on each individual probability p_i. A researcher may also care about functionals of the belief vector, such as a mean or another linear index; bounds on these are obtained from the same identified set $P_S(r)$.

This paper uses the term *informativeness* in an operational sense. A rule is more informative for a given design objective if the observed report leads to narrower coordinate bounds or narrower mean and functional bounds. These inferential gains must be weighed against implementation costs: exact-match payment probabilities, cognitive simplicity, computational burden, and robustness to risk aversion.

The paper's main analytical contribution is for squared-distance frequency scoring. Under this rule, a subject reports the feasible count vector closest to the expected count vector np. This gives a linear identified set and closed-form bounds for each p_i. Mean bounds and other linear functional bounds are then simple linear programs over the same region.

The paper compares this rule to two other count-loss mechanisms along a spectrum of analytical tractability. Frequency-guessing scoring, due to Schlag and Tremewan (2021), is the known exact-match mechanism: it elicits multinomial modes, gives closed-form bounds, and is robust to risk aversion under a fixed-prize implementation. Manhattan-distance scoring has an exact identified set and sharp, semi-analytical coordinate bounds.

The contribution is not a new exact-match mechanism and not a general theory of scoring rules; Schlag and Tremewan already derive belief bounds for the frequency-guessing rule. It is, first, the characterizations themselves: a closed-form identified set and coordinate bounds for squared-distance scoring, the centerpiece result, and sharp semi-analytical bounds for Manhattan distance, alongside the known frequency-guessing bounds. These three characterizations are not separate facts: a single structural observation—that an optimal report solves a separable resource-allocation problem—organizes them, and a classical convexity criterion then says which rules admit closed-form bounds and which only semi-analytical ones. It is, second, the contingent decision rule those characterizations make rigorous. No rule is uniformly best: squared-distance scoring is most informative when the subject's beliefs are concentrated on a few categories and frequency-guessing scoring when beliefs are balanced, with a crossover as concentration varies—first established analytically, where the two closed-form rules' average coordinate widths cross, then quantified and extended to Manhattan distance by the design comparison. Manhattan is rarely the most informative rule, but its bounds vary little with the regime, making it the robust choice when belief concentration cannot be anticipated. Partial identification is the lens that holds these together—each scoring rule maps an observed report to an identified set of latent beliefs—and is what lets the comparison be posed as a sharp, finite-sample question rather than an informal one. Two further count-loss rules, Hamming and Chebyshev distance, mark the boundary of the approach; Section 5 discusses them as rules whose identified sets are exact but resist tractable sharp bounds.

Related literature. The paper connects to, and should be read against, four literatures.

The closest is the work on *frequency guessing*. The frequency-guessing rule studied here is due to Schlag and Tremewan (2021): the subject reports a count vector and wins a fixed prize if it matches the realized counts. Schlag and Tremewan already establish, for this rule, the three steps the present paper relies on—the optimal report is a multinomial mode, the beliefs consistent with a report form a region of the simplex, and that region yields bounds on individual probabilities and on means and variances through linear programs. That inferential program, for this one rule, is theirs; the frequency-guessing results in Section 3 restate it so the comparison is self-contained. The contribution here is to carry the program beyond the rule for which it is known—to squared-distance scoring, where it delivers a closed-form identified set and closed-form coordinate bounds, and to Manhattan distance—and to turn the collection of rules into a comparison. Schlag and Tremewan also show that, for a binary event, frequency guessing is the most precise rule eliciting a single frequency; the question here is the different one of which rule is sharpest across belief regimes, where no rule wins outright.

A second literature is the theory of *elicitability*. The contrast drawn here between the mean-oriented squared-distance rule and the median-oriented Manhattan rule is an instance of a classical correspondence: squared-error loss elicits the mean and absolute-error loss elicits the median (Gneiting, 2011), and, more generally, a functional of a distribution is elicitable exactly when its level sets are convex (Lambert et al., 2008). That correspondence is not a contribution of this paper; its use is. The reports studied here are integer count vectors generated against a multinomial law, and each rule is inverted into an identified set of beliefs rather than evaluated as a point forecast. The frequency-guessing rule sits at the edge of the correspondence: it elicits the multinomial mode, and the mode is not directly elicitable for general distributions—it has the maximal elicitation complexity of the full distribution (Lambert et al., 2008) and is not elicitable at all in the continuous case (Heinrich, 2014). What makes mode elicitation possible here is the discreteness of the count outcome space, which renders the relevant identified set a polytope.

A third literature concerns what a coarse belief report identifies. A point forecast does not reveal whether it reports the mean, median, or mode of the underlying distribution (Gneiting, 2011; Engelberg et al., 2009), and Engelberg et al. (2009) accordingly bound the means, medians, and modes of forecasters' subjective distributions from their coarse interval-probability reports. Set-valued inference about a belief distribution from a coarse report is, in that sense, not new; the difference here is the identifying restriction. Engelberg, Manski, and Williams take the reported interval probabilities at face value, whereas the identified set $P_S(r)$ is cut out by the scoring rule's optimality condition—the report is informative only because it was incentive-generated. The count format itself is also established: survey research elicits subjective probability distributions directly (Manski, 2004), often by having respondents allocate a fixed number of tokens or beans across bins (Delavande et al., 2011), which is a frequency report in all but name.

A fourth literature is the experimental study of elicitation methods, surveyed in Schlag et al. (2015), Schotter and Trevino (2014), and Charness et al. (2021). Two of its recurring concerns are robustness to risk attitudes—direct monetary scoring rules distort reports for risk-averse subjects in ways that depend on stakes and hedging opportunities (Armantier and Treich, 2013)—and behavioural simplicity. Trautmann and van de Kuilen (2015) run a "horse race" among elicitation mechanisms on the accuracy of the beliefs they recover, and Offerman et al. (2009) keep a standard scoring rule but correct the elicited report ex post for risk attitudes and probability weighting; both aim to recover a point belief, by mechanism choice or by calibration. The exercise here differs in object: it holds optimal reporting fixed and compares count-loss rules by the sharpness of the belief *set* each report identifies, with no calibration step. That comparison is conditional on optimal reporting, an assumption that can fail behaviourally—Danz et al. (2022) find that the binarized scoring rule, a leading risk-robust mechanism, produces systematically centre-biased reports. The informativeness ranking obtained here is thus an upper bound on what a rule delivers, not a behavioural prediction; Section 5 returns to this as a scope condition.

2 Setup

The environment is a finite belief-elicitation problem. There are k possible outcomes, and the subject has latent beliefs

$$p = (p_1, \ldots, p_k) \in \Delta^k, \quad p_i \geq 0, \quad \sum_{i=1}^{k} p_i = 1.$$

The elicitation task asks about $n \in \mathbb{N}$ independent repetitions. If the subject's beliefs are p, the realized count vector is

$$\omega = (\omega_1, \ldots, \omega_k) \sim \mathrm{Mult}(n, p).$$

The set of feasible count vectors is

$$\Omega = \left\{ \omega \in \mathbb{N}_0^k : \sum_{i=1}^{k} \omega_i = n \right\},$$

and the multinomial probability mass function is

$$\Pr_p(\omega) = \frac{n!}{\omega_1! \cdots \omega_k!} \prod_{i=1}^{k} p_i^{\omega_i}.$$

The subject reports another feasible count vector

$$r = (r_1, \dots, r_k) \in \Omega.$$

Definition 1 (Frequency-report scoring rule)**.** A frequency-report scoring rule is a function

$$S : \Omega \times \Omega \to \mathbb{R},$$

where $S(r, \omega)$ is the score assigned to report r when the realized count vector is ω. The paper studies rules that reward reports close to realized counts. A convenient form is

$$S_D(r, \omega) = a - bD(r, \omega), \quad b > 0,$$

where $D : \Omega \times \Omega \to \mathbb{R}_+$ is a count-error index. The function D need not be a metric: squared distance is a divergence-style count loss, not a metric, because it fails the triangle inequality.

Two clarifications of terminology are in order. A *frequency report* $r \in \Omega$ is a vector of counts—how many of the n realizations fall in each category—not a vector of relative frequencies; dividing by n recovers the latter. And these are not *proper* scoring rules in the sense of Savage (1971) and Gneiting and Raftery (2007): they are not designed to make a truthful report of the belief vector p optimal, and generally do not. As the next section shows, the optimal report is instead a functional of the multinomial sampling distribution that p induces—its mode, its mean, or its coordinate-wise medians, depending on the rule—so recovering p calls for the inverse, partial-identification step developed below.

The constants a and b do not matter for expected-score rankings as long as $b > 0$. They matter for implementation. If scores are paid directly as money and nonnegative payments are desired, one can choose

$$a \geq b \max_{r, \omega \in \Omega} D(r, \omega).$$

How the score is implemented as a payment affects robustness to the subject's risk attitude; the paper's discussion of risk aversion and implementation takes up this question separately.

Definition 2 (Optimal reports)**.** Under risk neutrality, the optimal-report correspondence induced by a scoring rule S is

$$R_S(p) = \arg\max_{r \in \Omega} \mathbb{E}_p[S(r, \omega)].$$

Because Ω is finite, $R_S(p)$ is nonempty.

Risk neutrality is the maintained assumption throughout the body of the paper; the discussion of risk aversion and implementation separately shows how the analysis extends to risk-averse expected-utility subjects.

This is the forward incentive object. It answers the question: if the subject's latent beliefs are p, which reports are optimal? The econometric object is the inverse question. After observing r, the researcher asks which beliefs could have generated that report.

Definition 3 (Identified set)**.** For an observed report $r \in \Omega$, the identified set induced by S is

$$P_S(r) = \{p \in \Delta^k : r \in R_S(p)\}.$$

Ties are included: if r is one of several optimal reports under p, then $p \in P_S(r)$.

We call $P_S(r)$ the *mechanism-induced identified set*: the identifying restriction is the optimal-report condition $r \in R_S(p)$ imposed by the scoring-rule mechanism, not a sampling distribution over repeated observations. By construction $P_S(r)$ is exactly the set of beliefs consistent with that restriction.

The most direct summaries of $P_S(r)$ are coordinate-wise identification bounds:

$$\underline{p}_{S,i}(r) = \inf_{p \in P_S(r)} p_i, \quad \overline{p}_{S,i}(r) = \sup_{p \in P_S(r)} p_i,$$

with interval

$$P_{S,i}(r) = [\underline{p}_{S,i}(r), \overline{p}_{S,i}(r)].$$

When the scoring rule is clear from context, the subscript S is omitted.

Many experiments require more than coordinate-by-coordinate probability bounds. If outcomes have numerical values $x_1, \dots, x_k$, the latent mean implied by beliefs p is

$$\mu(p; x) = \sum_{i=1}^{k} p_i x_i.$$

This could represent an expected outcome, an expected treatment share, a predicted group average, or another linear index of the belief distribution. The sharp mean bounds after observing r are

$$\underline{\mu}_S(r; x) = \inf_{p \in P_S(r)} \sum_{i=1}^{k} p_i x_i, \quad \overline{\mu}_S(r; x) = \sup_{p \in P_S(r)} \sum_{i=1}^{k} p_i x_i.$$

The corresponding mean interval is

$$[\underline{\mu}_S(r; x), \overline{\mu}_S(r; x)].$$

These bounds generally require optimizing over the full joint region $P_S(r)$. Coordinate intervals alone do not generally give sharp bounds for a mean, because they ignore dependence across coordinates imposed by the simplex and the scoring rule.

The design comparison later summarizes the informativeness of a rule using coordinate widths,

$$\overline{p}_{S,i}(r) - \underline{p}_{S,i}(r),$$

and functional widths,

$$\overline{\mu}_S(r; x) - \underline{\mu}_S(r; x).$$

Exact-match payment probability is reported only for the frequency-guessing rule, as an implementation diagnostic of that fixed-prize mechanism; it is not a cross-rule comparison metric.

The next section applies this setup to the three headline rules—squared-distance, frequency-guessing, and Manhattan—beginning with a structural lemma that links them.

3 Frequency-Report Scoring Rules

This section applies the setup to three frequency-report scoring rules. They span a spectrum of analytical tractability. Squared-distance and frequency-guessing scoring have closed-form identified sets and closed-form coordinate bounds. Manhattan-distance scoring has an exact identified set and sharp coordinate bounds in semi-analytical form. A single structural observation, developed first, accounts for that spectrum—and for the two further rules, in Section 5, that fall outside it.

3.1 Optimal Reports as a Resource-Allocation Problem

The three rules share a structure that determines what a report reveals. For each, an optimal report minimizes a *separable* objective over the feasible count vectors,

$$R_S(p) = \arg\min_{r\in\Omega} \sum_{i=1}^{k} \ell(r_i; p_i),$$

for a per-coordinate cost ℓ. For squared-distance and Manhattan scoring the expected loss is itself additively separable, and ℓ is the expected per-coordinate loss. Frequency-guessing scoring is not an additively separable loss—the multinomial probability is a product—but maximizing $\Pr_p(\omega = r)$ is equivalent to minimizing $\sum_i \{\log(r_i!) - r_i \log p_i\}$, so it has the same form, with $\ell(t; p_i) = \log(t!) - t \log p_i$.

This is the integer *separable resource-allocation problem*: distribute n units across k categories under separable costs. Its solution structure is classical (Fox, 1966; Federgruen and Groenevelt, 1986; Ibaraki and Katoh, 1988).

Lemma 1 (Single-transfer characterization)**.** *Suppose an optimal report minimizes a separable objective $\sum_i \ell(r_i; p_i)$ over Ω, and each $\ell(\cdot; p_i)$ is* discrete-convex*: its successive differences $\ell(t+1; p_i) - \ell(t; p_i)$ are nondecreasing in t. Then r is optimal if and only if no single-count transfer lowers the objective,*

$$\ell(r_i + 1; p_i) - \ell(r_i; p_i) \;\geq\; \ell(r_j; p_j) - \ell(r_j - 1; p_j) \qquad \forall\, i, j \text{ with } r_j > 0,$$

and the identified set $P_S(r)$ is exactly the set of beliefs satisfying these $O(k^2)$ inequalities.

Necessity of the no-transfer condition is immediate: a transfer produces a feasible competing report. Sufficiency is a monotone-path exchange argument—any competing report is reached by single-count transfers along which discrete convexity makes each step at least as costly as the corresponding transfer evaluated at r; it is proved in Appendix A.1. That the equivalence holds precisely when the per-coordinate costs are discrete-convex is the classical greedy-optimality characterization for separable resource allocation (Fox, 1966; Federgruen and Groenevelt, 1986), with discrete convexity the standard diminishing-differences condition (Murota, 2003).

The three headline rules satisfy the hypothesis: the squared-distance cost is quadratic in r_i, the Manhattan cost $\mathbb{E}_p|r_i - \omega_i|$ inherits discrete convexity from the absolute value, and the frequency-guessing cost $\log(r_i!) - r_i \log p_i$ is discrete-convex because $\log(t!)$ has nondecreasing differences. Each rule's characterization below is therefore an instance of Lemma 1; the rules differ only in the shape of the single-transfer inequalities. When those inequalities are linear in

p—squared-distance and frequency-guessing—the identified set is a polytope and the coordinate bounds are closed form; when they are monotone but transcendental in p—Manhattan—the identified set is still exact and the bounds are sharp, but only semi-analytically. The lemma's reach is not confined to these three: other separable count losses are governed by it in the same way—for instance asymmetric-absolute losses, which elicit quantiles other than the median, and the ranked probability score, a squared distance on cumulative counts—so the headline rules are representatives of a wider tractable family rather than a closed list.

The lemma also marks the limits of the approach. Its two hypotheses—separability and discrete convexity—are exactly what a count loss can fail, and Section 5 takes up the two rules that do: Hamming distance is separable but its per-coordinate cost is not discrete-convex, and Chebyshev distance is not separable. The optimization content of the lemma is classical; what this paper adds is the observation that incentivized frequency-report elicitation *is* a resource-allocation problem of this kind, and the inversion of the resulting optimal-report correspondence into an identified set of latent beliefs.

3.2 Squared-Distance Scoring

The main analytical rule is squared-distance frequency scoring. It pays less when the reported count vector is far from the realized count vector in squared Euclidean distance:

$$S_Q(r, \omega) = a - b\|r - \omega\|_2^2, \quad b > 0.$$

This rule is connected in spirit to quadratic scoring rules, but it is not a standard probability score. It is a frequency-report count-loss rule: subjects report counts, and the penalty is the squared difference between reported and realized counts. Write P_Q for the identified set induced by this rule.

Given beliefs p, the expected count vector is np. The key forward fact is that squared-distance scoring asks the subject to report the feasible integer count vector closest to np. The inverse question is therefore geometric: for which beliefs is the observed report r the closest feasible count vector to np?

Proposition 1 (Squared distance: reports, regions, and bounds)**.** *For every $p \in \Delta^k$,*

$$R_{S_Q}(p) = \arg\min_{s \in \Omega} \|s - np\|_2^2.$$

For an observed report $r \in \Omega$, the identified set is

$$P_Q(r) = \left\{ p \in \Delta^k : n(p_i - p_j) \le r_i - r_j + 1 \quad \forall i, j \text{ with } r_j > 0 \right\}.$$

Let

$$m(r) = |\{j : r_j > 0\}|$$

be the number of categories with positive reported counts. The sharp coordinate intervals are

$$P_{Q,i}(r) = \begin{cases} \left[0, \dfrac{m(r)}{n(m(r)+1)}\right], & r_i = 0, \\[2ex] \left[\dfrac{r_i - 1}{n} + \dfrac{1}{nk}, \dfrac{r_i + 1}{n} - \dfrac{1}{n\,m(r)}\right], & r_i > 0. \end{cases}$$

Projection ties are included.

The inequalities compare the observed report r with reports obtained by moving one count from a positive reported category j to another category i. If all such single-count moves weakly increase squared distance from np, then r is a closest feasible report. Thus $P_Q(r)$ is a polytope in the belief simplex—a region cut out by finitely many linear inequalities. The proof is in Appendix A.2.

Probability and mean bounds. The coordinate bounds are closed form. For numerical outcomes $x_1, \ldots, x_k$, sharp mean bounds are obtained by optimizing over the same polytope:

$$\min_{p \in P_Q(r)} \sum_{i=1}^{k} p_i x_i \quad \text{and} \quad \max_{p \in P_Q(r)} \sum_{i=1}^{k} p_i x_i.$$

These are linear programs. For coordinate probabilities, the displayed intervals give the corresponding linear-program solutions explicitly. This is the main practical advantage of squared-distance frequency scoring: the same transparent region gives both probability bounds and bounds on expected values.

Practical interpretation. Squared distance is mean-oriented: it links a report r to beliefs for which r is the integer projection of np—the count-report counterpart of squared-error loss eliciting the mean (Gneiting, 2011). Compared with exact matching, the payment varies smoothly with distance rather than only with exact equality. The main implementation cost is that direct monetary squared-distance payments generally require risk neutrality unless the score is implemented probabilistically, as discussed below.

3.3 Frequency-Guessing Scoring

The second closed-form rule is the frequency-guessing mechanism of Schlag and Tremewan (2021). Its count-error index is the count-vector discrete metric,

$$D_0(r, \omega) = \mathbf{1}\{r \neq \omega\},$$

so, in the form $S_D = a - bD$ of Section 2, the score is

$$S_0(r, \omega) = a - b\,\mathbf{1}\{r \neq \omega\}, \quad b > 0.$$

Equivalently, $S_0(r, \omega) = (a - b) + b\,\mathbf{1}\{r = \omega\}$: the subject receives an extra b whenever the realized count vector equals the report. This fixed-prize reading is useful for implementation and risk aversion; the two formulations induce the same optimal reports. Write R_0 and P_0 for the report correspondence and identified set induced by this rule.

The mechanism is simple: a subject with beliefs p maximizes the probability of an exact match,

$$\Pr_p(\omega = r) = \frac{n!}{r_1! \cdots r_k!} \prod_{i=1}^{k} p_i^{r_i}.$$

Thus the forward object is the multinomial mode set. The inverse object is the set of beliefs for which the observed report is one of those modes.

Proposition 2 (Frequency guessing: reports, regions, and bounds)**.** *Let $r \in \Omega$, and allow ties in the multinomial mode. Then r is optimal under frequency-guessing scoring if and only if r is a multinomial mode:*

$$R_0(p) = \arg\max_{s \in \Omega} \Pr_p(\omega = s).$$

The identified set is

$$P_0(r) = \left\{ p \in \Delta^k : r_j p_i \leq (r_i + 1) p_j \quad \forall i, j \text{ with } r_j > 0 \right\}.$$

For each coordinate, the sharp identification interval is

$$P_{0,i}(r) = \left[\frac{r_i}{n + k - 1}, \frac{r_i + 1}{n + 1} \right].$$

The inequalities say that no single-count transfer from a reported category j to another category i can increase the multinomial probability of the report. Weak inequalities include ties. The formula also handles boundaries: if $r_i = 0$, the lower bound is 0; if $r_i = n$, the upper bound is 1. These are the frequency-guessing bounds in Schlag and Tremewan (2021, Proposition 1); they are restated here to make the informativeness comparison self-contained. The proof is in Appendix A.3.

Probability and mean bounds. The frequency-guessing rule gives closed-form coordinate intervals. For linear functionals such as means, sharp bounds use the full joint region, not just the coordinate intervals:

$$\min_{p \in P_0(r)} \sum_{i=1}^{k} p_i x_i \quad \text{and} \quad \max_{p \in P_0(r)} \sum_{i=1}^{k} p_i x_i.$$

These are linear programs because $P_0(r)$ is defined by linear inequalities. Thus frequency-guessing scoring is analytically convenient for individual probability bounds, while mean inference still requires using the joint mode region.

Practical interpretation. Frequency-guessing scoring is transparent and robust to risk aversion because it is a fixed-prize event. Its implementation cost is that the probability of an exact match can be small when n is large or the outcome space is rich. It is a prominent known rule in the comparison, not a new mechanism introduced here.

3.4 Comparing the Two Closed-Form Rules

The closed-form coordinate bounds of Propositions 1 and 2 can be compared directly, before any simulation. For a rule S and report r, call the mean of the coordinate widths $\overline{p}_{S,i}(r) - \underline{p}_{S,i}(r)$ over the k coordinates the *average coordinate width.*

Corollary 1 (Average coordinate width of the closed-form rules)**.** *Fix $n \geq 2$ and $k \geq 2$, and for a report r let $m = m(r)$ be the number of categories with positive reported counts.*

1. *The frequency-guessing average coordinate width equals the constant*

$$W_0(n,k) = \frac{(k-1)(2n+k)}{k(n+1)(n+k-1)},$$

   the same for every report r.

2. *The squared-distance average coordinate width depends on* r *only through* m*:*

$$W_Q(m;n,k) = \frac{1}{nk}\left[2m - 1 - \frac{m}{k} + \frac{m(k-m)}{m+1}\right].$$

3. *The two widths cross:* $W_Q(1;n,k) < W_0(n,k) < W_Q(k;n,k)$.

The proof—elementary algebra on the two coordinate-bound formulas—is in Appendix A.4.

Parts (1) and (2) together say something clean: the frequency-guessing average width is a constant, while the squared-distance average width is a function of m alone, so the comparison between the two closed-form rules is decided entirely by $m = m(r)$, the number of categories the report touches. Part (3) shows that comparison is not one-sided—the squared-distance average width falls below the frequency-guessing constant at the most concentrated report and rises above it at $m = k$—so which rule gives the narrower average coordinate bound genuinely turns on how concentrated the report is. The design comparison of the next section ties this report-level contrast to belief concentration, quantifies it, and extends it to Manhattan distance.

3.5 Manhattan-Distance Scoring

Unlike the two closed-form rules, Manhattan-distance scoring has no closed-form coordinate bounds; its optimal-report correspondence and identified set are nonetheless characterized exactly, and its sharp bounds reduce to a one-dimensional computation. The Manhattan-distance rule is

$$S_M(r,\omega) = a - b\sum_{i=1}^{k} |r_i - \omega_i|, \quad b > 0.$$

It rewards total absolute count accuracy rather than squared count accuracy, and its inferential interpretation is median-oriented rather than mean-oriented—the count-report counterpart of absolute-error loss eliciting the median (Gneiting, 2011).

For category i the realized count has binomial marginal $W_i \sim \mathrm{Bin}(n, p_i)$; write $F_i(t;n,p_i) = \Pr(W_i \le t)$ for its distribution function, with the conventions $F_i(-1;n,p_i) = 0$ and $F_i(n;n,p_i) = 1$. By linearity of expectation the expected Manhattan loss is separable across categories, $\mathbb{E}_p \sum_i |r_i - \omega_i| = \sum_i \mathbb{E}_p|r_i - W_i|$, while the simplex constraint $\sum_i r_i = n$ couples the coordinates.

Proposition 3 (Manhattan distance: optimal reports and identified set)**.** *Optimal reports minimize the separable expected Manhattan loss,*

$$R_M(p) = \operatorname*{arg\,min}_{r\in\Omega} \sum_{i=1}^{k} \mathbb{E}_p|r_i - W_i|,$$

and r is optimal if and only if no single-count transfer lowers that loss, equivalently

$$F_i(r_i; n, p_i) \geq F_j(r_j - 1; n, p_j) \qquad \forall\, i, j \text{ with } r_i < n,\ r_j > 0.$$

This condition is sufficient as well as necessary, so the identified set is exactly

$$P_M(r) = \Big\{ p \in \Delta^k :\ F_i(r_i; n, p_i) \geq F_j(r_j - 1; n, p_j)\ \ \forall\, i, j \text{ with } r_i < n,\ r_j > 0 \Big\}.$$

Equivalently, $p \in P_M(r)$ if and only if some threshold $c \in [0, 1]$ satisfies

$$F_i(r_i - 1; n, p_i) \leq c \leq F_i(r_i; n, p_i) \qquad \forall\, i.$$

Each coordinate loss $\mathbb{E}_p|t - W_i|$ is discrete-convex in the reported count t—its successive increments are nondecreasing—so Lemma 1 applies and $P_M(r)$ is the exact identified set rather than an outer approximation. Weak inequalities include ties, and there is no receiver constraint when $r_i = n$ nor sender constraint when $r_j = 0$. The proof is in Appendix A.5.

Identification bounds. For $k = 2$, write $r = (t, n - t)$; the Manhattan loss is $2|t - W|$ with $W \sim \text{Bin}(n, p_1)$, and the report identifies the binomial-median interval

$$P_{M,1}(r) = \{p_1 \in [0, 1] : F(t - 1; n, p_1) \leq 1/2 \leq F(t; n, p_1)\}.$$

This binary case is clean.

For $k > 2$ the threshold representation of Proposition 3 drives the bounds. At a fixed threshold c the inequalities $F_i(r_i - 1; n, p_i) \leq c \leq F_i(r_i; n, p_i)$ invert, coordinate by coordinate, to a box of beliefs through inverse binomial CDFs; intersecting that box with the simplex gives the feasible beliefs at that threshold. Each sharp coordinate bound is then the optimum, over $c \in [0, 1]$, of a function that is unimodal in c, so it is the solution of a single monotone scalar equation. The coordinate bounds are therefore sharp and semi-analytical: not closed form, because the inverse binomial CDF is not elementary, but exact and one-dimensional rather than grid-limited. The sharp mean bounds are obtained by the same one-dimensional search over the threshold.

Practical interpretation. Manhattan distance is useful when a median-type count prediction is substantively appropriate. It is also useful as a contrast: changing the distance criterion changes the inferred identified set. The multi-category Manhattan bounds are sharp and reduce to a one-dimensional threshold computation; they are less transparent than the closed-form squared-distance bounds but are exact, not approximations.

4 Informativeness: A Design Comparison

The analytical regions are useful because they tell an experimenter what can be inferred after observing a report. This section reports a design comparison generated by the reproducible script `scripts/design_efficiency.py`. The exercise draws latent beliefs

$$p \sim \text{Dirichlet}(\alpha, \ldots, \alpha)$$

and, for each of the three rules with tractable sharp bounds—frequency-guessing, squared-distance, and Manhattan distance—computes an optimal report $r_S(p)$ and the bounds induced by $P_S(r_S(p))$. Hamming and Chebyshev distance fall outside this comparison; Section 5 takes them up as count-loss rules whose identified sets are exact but resist tractable sharp bounds. The final run uses 5,000 belief draws for each cell in

$$n \in \{5, 10, 20, 50\}, \qquad k \in \{2, 3, 5, 10\}, \qquad \alpha \in \{0.1, 0.3, 1, 3, 10\}.$$

The Dirichlet parameter α sets belief concentration: small α draws sparse beliefs near a vertex of the simplex, large α draws balanced, interior beliefs. For each rule the comparison records two headline inferential targets—the average width of the coordinate bounds, and the width of the bound on the linear functional $\sum_i p_i x_i$ with outcome values $x_i = (i-1)/(k-1)$. Manhattan coordinate bounds are sharp, computed by a threshold root-find; Manhattan mean bounds are threshold-computed with tolerance 10^{-4}.

The comparison is conditional on the theoretical characterizations above. It is not evidence of incentive compatibility, and it is not an optimality theorem. It plays three roles: it confirms the analytic crossover between the two closed-form rules established in Section 3, quantifies that crossover through win shares and regret, and extends the comparison to Manhattan distance, whose lack of a closed-form bound rules out a comparable analytic treatment. It is, in short, a diagnostic for the practical question: which rule gives a researcher tighter inference for the object of interest?

The ranking is governed by belief concentration. Corollary 1 showed, from the closed-form bounds alone, that which of the two closed-form rules gives the narrower average coordinate bound turns on report concentration—the number m of categories with positive reported counts. The design comparison shows that report concentration tracks belief concentration, so the crossover is governed by how concentrated the subject's beliefs are. Averaged over the whole grid the three rules barely differ—mean coordinate width 0.102, 0.104, and 0.102 for frequency-guessing, squared-distance, and Manhattan scoring—because each rule's favourable and unfavourable regimes cancel. Table 1 reports the win shares—the share of belief draws on which a rule gives the narrowest interval—broken out by α.

	Average-coordinate			Linear-mean		
α	Guessing	Squared	Manhattan	Guessing	Squared	Manhattan
0.1	0.08	0.84	0.07	0.08	0.83	0.10
0.3	0.28	0.58	0.14	0.22	0.60	0.18
1	0.76	0.16	0.08	0.58	0.27	0.15
3	0.98	0.01	0.01	0.83	0.13	0.04
10	1.00	0.00	0.00	0.92	0.06	0.01

Table 1: Win share by belief concentration α: the share of belief draws on which a rule gives the narrowest interval, averaged over all n and k at each α. Squared-distance scoring wins for sparse beliefs (small α); frequency-guessing scoring for balanced beliefs (large α).

Squared-distance scoring is the most informative rule when beliefs are sparse and frequency-guessing scoring when they are balanced, with a crossover near $\alpha = 1$. At $\alpha = 0.1$ squared-distance gives the narrowest average-coordinate interval on 0.84 of draws; by $\alpha \geq 3$ frequency-guessing does so on essentially every draw. The linear-mean target shows the same crossover. The number of categories k and the number of trials n are second-order, as Table 3 in Appendix B documents: for the coordinate target the ranking is nearly flat in both, while for the mean target larger n mildly favours squared-distance and the binary case $k = 2$ favours frequency-guessing. The worst-coordinate target follows the same pattern shifted in squared-distance's favour—squared-distance gives the narrowest maximum-coordinate interval on a majority of draws for every $\alpha \leq 3$. The ranking is robust to the choice of outcome vector and is unaffected by optimal-report ties, which are negligible under the Dirichlet draws; Appendix B reports both checks.

The crossover has a mechanism in the projection interpretation of squared-distance scoring. A subject reports the integer count vector closest to np; when beliefs are sparse, np lies near a vertex of the simplex, the feasible reports around it are extreme count vectors, and the projection polytope is tight, so the induced bounds are narrow. As beliefs become balanced, np moves into the interior, the projection polytope widens, and frequency-guessing scoring—which elicits the multinomial mode—becomes the more informative rule.

The cost of the wrong rule, and Manhattan as a hedge. Win shares count only first places. Cell-best regret—a rule's interval width minus the narrowest width available in the same cell—measures how much precision is forfeited by not using the best rule. Table 2 reports regret by α.

	Average-coordinate			Linear-mean		
α	Guessing	Squared	Manhattan	Guessing	Squared	Manhattan
0.1	0.032	0.002	0.011	0.047	0.002	0.013
0.3	0.016	0.005	0.007	0.024	0.005	0.009
1	0.003	0.012	0.007	0.006	0.013	0.008
3	0.000	0.019	0.011	0.001	0.020	0.012
10	0.000	0.023	0.014	0.000	0.024	0.016

Table 2: Mean cell-best regret by belief concentration α: a rule's interval width minus the smallest width among the three rules, averaged over all n and k at each α. Lower is better. Frequency-guessing and squared-distance scoring each swing between near-zero and substantial regret; Manhattan's regret is nearly flat.

Frequency-guessing and squared-distance scoring each have a regime of near-zero regret and a regime in which regret is substantial. Frequency-guessing's average-coordinate regret runs from 0.000 at $\alpha = 10$ to 0.032 at $\alpha = 0.1$; squared-distance's runs the other way, reaching 0.024 on the mean target at $\alpha = 10$. Against typical interval widths near 0.10 to 0.15, the worst-regime values are a quarter or more of an interval width—a real loss of precision from choosing the rule that does not match the belief regime. Manhattan distance behaves differently. It is never the narrowest rule, but its regret is nearly constant in α—between 0.007 and 0.016 at every concentration and for both targets—so it is never far from the best rule either. Manhattan is therefore the regime-robust choice: a researcher who cannot anticipate how concentrated subjects' beliefs are minimises worst-case regret by using it, whereas committing in advance to frequency-guessing or squared-distance scoring risks the unfavourable regime. This is the precise sense in which Manhattan is a low-regret rule: it hedges the belief-concentration uncertainty to which the other two rules are exposed.

Payment probability. The frequency-guessing rule has a fixed-prize implementation, which is attractive under risk aversion. Its implementation cost is that the exact-match payment probability can be small. In the final run, the average exact-match probability is high for very sparse binary beliefs, but it falls sharply as n, k, and belief balance increase; for instance, it is essentially zero in the $n = 50, k = 10, \alpha \in \{1, 3, 10\}$ cells. Thus narrow bounds under frequency-guessing scoring should be weighed against the chance that the subject rarely wins the fixed prize.

The comparison thus delivers a contingent recommendation, not a single winner: the most informative rule is the one matched to the anticipated belief regime and inferential target, a choice the discussion turns into practical guidance.

5 Discussion

The paper studies frequency-report scoring rules as inferential devices. The report r is not the belief vector. It is evidence about p, and the scoring rule determines how informative that evidence is. The relevant chain is therefore

$$S \;\rightarrow\; R_S(p) \;\rightarrow\; P_S(r) \;\rightarrow\; \big[\underline{p}_{S,i}(r), \overline{p}_{S,i}(r)\big].$$

The design comparison shows that the most informative rule depends on how concentrated the subject's beliefs are; three practical recommendations follow. First, use frequency-guessing scoring when the subject is expected to spread probability fairly evenly across the outcomes, with no single outcome carrying most of the probability. It is the most informative rule in that regime, has closed-form coordinate bounds, and is the only rule robust to risk aversion under direct payment; its implementation cost is that exact-match payment probabilities can be low away from concentrated reports.

Second, use squared-distance frequency scoring when the subject is expected to place most of the probability on one or a few outcomes. It is the most informative rule in that regime—for the average-coordinate, worst-coordinate, and mean targets alike—and it offers a transparent projection rule, a linear identified set, closed-form coordinate bounds, and linear-program bounds for any linear functional.

Third, use Manhattan distance when it cannot be anticipated in advance whether the subject's beliefs will be concentrated on a few outcomes or spread across many. Its identified set is exact and its coordinate bounds are sharp—semi-analytical, reducing to a one-dimensional threshold computation rather than a closed form. Manhattan rarely gives the single narrowest interval, but the design comparison shows its regret is nearly constant across belief concentrations: it is never far from the best rule, and so minimizes worst-case regret for a researcher who does not know the regime. Its inferential interpretation is also median-oriented, which suits settings where a median-type count prediction is substantively appropriate.

5.1 Risk Aversion and Implementation

The body of the paper assumes risk neutrality. Whether that assumption can be relaxed depends on how the scoring rule is implemented as a payment. The distinction is between paying the score directly as money and using the score to determine the probability of a fixed prize.

Direct monetary scores. If the subject is paid the score itself, the subject solves

$$\arg\max_{r\in\Omega} \mathbb{E}_p[u(S(r,\omega))].$$

For the frequency-guessing rule this does not change incentives. The payment is a fixed prize B if $r = \omega$ and zero otherwise, so

$$\mathbb{E}_p[u] = u(0) + \Pr_p(\omega = r)\{u(B) - u(0)\},$$

and any strictly increasing u preserves the same optimal reports as maximizing the exact-match probability. Direct monetary distance scores do not have this robustness: squared-distance and Manhattan distance generate several possible payment levels, so nonlinear utility can change the ranking of reports. This is the standard risk-aversion problem for monetary scoring rules (Armantier and Treich, 2013; Schlag et al., 2015). One response in the literature keeps the scoring rule unchanged and corrects the elicited report ex post for the subject's risk attitude and probability weighting (Offerman et al., 2009); the route taken here is instead to choose a rule, or an implementation, under which no such correction is needed.

Probability implementation. The distance rules can instead be implemented as a binary lottery. Let $q(r,\omega) \in [0,1]$ be the score normalized into a probability, and pay a fixed prize B with probability $q(r,\omega)$ and zero otherwise. If the prize and outside payment are fixed, utility depends only on final money, and the randomization is objective and independent, then

$$\mathbb{E}_p[q(r,\omega)u(B) + (1 - q(r,\omega))u(0)] = u(0) + (u(B) - u(0))\,\mathbb{E}_p[q(r,\omega)],$$

so maximizing expected utility is equivalent to maximizing the expected normalized score, regardless of the curvature of u. A positive affine normalization of the bounded score into $[0,1]$ preserves the expected-score ranking of reports, so the body's risk-neutral analysis carries over unchanged. This is the binary-lottery, or binarized-scoring-rule, technique (Roth and Malouf, 1979; Berg et al., 1986; Karni, 2009; Hossain and Okui, 2013); the contribution here is to apply it, not to introduce it.

The algebra above establishes only *theoretical* robustness, and two qualifications bound what it buys in practice. First, the robustness is to risk aversion *within* expected utility. Subjects who depart from expected utility—for instance through probability weighting—can report non-truthfully even under the binary-lottery implementation, and the paper does not address such departures. Second, the implementation asks subjects to reason about a compound lottery layered on top of the count report, which partially offsets the cognitive simplicity that motivates frequency reports in the first place. The empirical record reflects this tension. Harrison et al. (2014) find that a binary-lottery procedure does induce approximately risk-neutral reporting in a subjective-probability elicitation task, evidence that the technique can deliver its theoretical promise. But Danz et al. (2022), testing the binarized scoring rule directly, find that informing subjects of the quantitative incentives induces systematic centre-biased misreporting—a failure they trace not to risk attitudes, which the binary lottery already neutralizes, but to the incentive information itself. The binary-lottery implementation should therefore be read as resolving the risk-attitude problem the body sets aside while leaving a separate, behavioural one open.

The implementation choice also interacts with a cost noted in the design comparison. The frequency-guessing prize is paid only on an exact match, and the exact-match probability falls sharply as n, k, and belief balance increase, so at larger designs the rule becomes a lottery with a vanishing chance of any payment. The frequency-guessing rule is thus robust to risk aversion even under direct payment but can become nearly unwinnable; the distance rules avoid that under the binary-lottery implementation, whose payment probability is the expected normalized score, at the cost of relying on expected utility.

5.2 Other Frequency-Report Scoring Rules and the Limits of the Approach

The three rules compared above do not exhaust the family of count losses, and the partial-identification approach is not a general theory of discrete scoring rules. Two further rules—Hamming and Chebyshev distance—are worth stating precisely, because they mark where the approach stops yielding tractable sharp bounds. They are, in fact, the two ways the structural lemma of Section 3 (Lemma 1) can fail: Hamming distance meets its separability hypothesis but not its discrete-convexity hypothesis, and Chebyshev distance fails separability itself.

Hamming distance. The Hamming rule $S_H(r, \omega) = a - b \sum_i \mathbf{1}\{r_i \neq \omega_i\}$ penalizes the number of categories whose reported count differs from the realized count. By linearity of expectation its expected loss is $\sum_i \{1 - \Pr(\mathrm{Bin}(n, p_i) = r_i)\}$, so an optimal report maximizes the separable objective $\sum_i \Pr(\mathrm{Bin}(n, p_i) = r_i)$ subject to $\sum_i r_i = n$, and $P_H(r)$ is the set of beliefs at which r attains that maximum—cut out by one expected-loss inequality per alternative report. For $k = 2$ the two coordinate indicators $\mathbf{1}\{r_i \neq \omega_i\}$ are equal, so the Hamming count-error index $\sum_i \mathbf{1}\{r_i \neq \omega_i\}$ is exactly twice the discrete metric $\mathbf{1}\{r \neq \omega\}$; the rule coincides with frequency-guessing scoring and inherits its closed-form bounds. For $k > 2$ it does not. Each term $\Pr(\mathrm{Bin}(n, p_i) = r_i)$ is unimodal but not discrete-concave in the reported count, so the exchange argument that makes single-count-transfer optimality sufficient for Manhattan distance fails here: the condition remains necessary but is no longer sufficient, because a transfer of two or more counts can raise the separable objective where no single-count transfer does. This failure is not confined to the boundary of the simplex. At $n = 3$, $k = 5$, and uniform beliefs $p = (1/5, \ldots, 1/5)$, for instance, the report $(3, 0, 0, 0, 0)$ admits no improving single-count transfer, yet $(0, 0, 1, 1, 1)$ yields a strictly higher expected score. The identified set $P_H(r)$ is therefore a non-convex set defined by polynomial inequalities, not a polytope. The modal box $\prod_i [r_i/(n+1), (r_i+1)/(n+1)]$ collects the beliefs at which every r_i is a mode of its binomial marginal; there the separable objective is maximized term by term, so r is optimal, and the modal box is a closed-form inner bound for $P_H(r)$, while the single-count-transfer inequalities give an outer bound. The sharp coordinate bounds lie between these two and require optimizing a coordinate directly over the non-convex set—a global optimization whose feasible region is cut out by one polynomial inequality for each of the $\binom{n+k-1}{k-1}$ feasible reports, a constraint count that grows combinatorially and exceeds 10^{10} at the largest cell of the design comparison ($n = 50$, $k = 10$). A direct feasibility check bore out the obstruction: computing the sharp Hamming bounds proved both prohibitively slow and numerically unreliable for $k = 10$, while the closed-form bounds of the other rules are immediate. Hamming distance is for this reason characterized here rather than included in the comparison.

Chebyshev distance. The Chebyshev rule $S_\infty(r,\omega) = a - b\max_i |r_i - \omega_i|$ penalizes the largest single-category count error. Unlike the other count losses, its expected loss does not separate across coordinates: the maximum couples the category errors within every realization, so there is no per-coordinate transfer condition to exploit. For $k = 2$ the two absolute errors $|r_i - \omega_i|$ are equal, so the count-error index $\max_i |r_i - \omega_i|$ is half of $\sum_i |r_i - \omega_i|$; the rule coincides with Manhattan scoring and inherits the binomial-median interval. For $k > 2$ neither half of the pipeline is tractable at scale. The non-separable objective admits no transfer or exchange argument, so no characterization of the optimal report beyond search is known: a certified optimal report must be found by comparing all $\binom{n+k-1}{k-1}$ feasible reports. The identified set, in turn, is cut out only by the full system of expected-loss inequalities, one per alternative report, with no transfer-polytope or threshold reduction—the same kind of non-convex semialgebraic set as $P_H(r)$, whose sharp bounds the Hamming feasibility check above already found intractable at $k = 10$. Chebyshev distance is therefore computable by enumeration for small n and k but, like Hamming, cannot be carried across the design grid; it is characterized here rather than included in the comparison.

These two rules are not pathological—their identified sets are well defined and finite to check—but they show that the closed-form and semi-analytical tractability of the headline rules is a property of those rules, not a generic feature of count-loss scoring.

5.3 An Open Empirical Question

The case for frequency reports rests partly on a behavioural premise this paper does not test: that subjects reason more readily in counts than in abstract probabilities, and find a distance-based payment more transparent than a probability score. That premise is taken here on the strength of the cited literature. A second, more basic premise maintained throughout the body is that subjects report optimally given the rule—a premise that, as the risk-aversion discussion above noted, the evidence on incentive-compatible mechanisms does not uniformly support (Danz et al., 2022). The informativeness comparison is therefore best read as a statement about the mechanism conditional on optimal reporting, not as a behavioural prediction. The partial-identification framework itself, however, suggests how both premises could be examined. If beliefs are induced by the experimenter—through draws from an urn of known composition, or a signal structure with a computable posterior—the true belief p is known, so an elicited report r can be checked against the identified set $P_S(r)$ it generates: under optimal reporting $P_S(r)$ contains p, and the empirical coverage rate, with the direction of any systematic miss, measures how far subjects depart from optimal reporting. Eliciting the same induced beliefs under a frequency-report rule and under a standard probability scoring rule isolates the effect of the report format; crossing that with direct versus binary-lottery payment isolates the effect of the implementation, which the risk-aversion analysis above predicts to matter only for subjects who are not risk neutral. Response times, incentivized comprehension questions, and the rate of weakly dominated reports would complement the coverage measure. Such a design would test, rather than presume, the behavioural premise on which the case for frequency reports partly rests—a premise separate from the present contribution, which is that different frequency-report scoring rules induce different identified sets, and those sets determine the informativeness of the mechanism.

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A Proofs

A.1 The Single-Transfer Characterization

Proof of Lemma 1. Write $\Delta\ell(t;q) = \ell(t+1;q) - \ell(t;q)$; discrete convexity of $\ell(\cdot;q)$ means $\Delta\ell(\cdot;q)$ is nondecreasing.

Necessity. For any i, j with $r_j > 0$ the report $r + e_i - e_j$ is feasible, and it changes the objective by $\Delta\ell(r_i;p_i) - \Delta\ell(r_j - 1;p_j)$. If r is optimal this change is nonnegative, which is the displayed inequality.

Sufficiency. Suppose r satisfies every displayed inequality, and let $s \in \Omega$ be any competing report. Since $\sum_i s_i = \sum_i r_i = n$, the surplus units $\sum_i (s_i - r_i)^+$ and the deficit units $\sum_j (r_j - s_j)^+$ are equal in number, say M. Reach s from r by M single-count transfers, each moving one unit from a coordinate still above its target s to one still below. A receiver coordinate then only rises, from r_i toward s_i, and a sender only falls, from r_j toward s_j; so every intermediate report is feasible, and at the step transferring a unit $j \to i$ the running counts satisfy $c_i \geq r_i$ and $c_j \leq r_j$. That step changes the objective by $\Delta\ell(c_i;p_i) - \Delta\ell(c_j - 1;p_j)$. Because $\Delta\ell(\cdot;p)$ is nondecreasing, $c_i \geq r_i$ gives $\Delta\ell(c_i;p_i) \geq \Delta\ell(r_i;p_i)$, and $c_j - 1 \leq r_j - 1$ gives $\Delta\ell(c_j - 1;p_j) \leq \Delta\ell(r_j - 1;p_j)$; so the step's change is at least $\Delta\ell(r_i;p_i) - \Delta\ell(r_j - 1;p_j)$, nonnegative by the displayed inequality for the pair (i, j). Summing over the M steps gives $\sum_i \ell(s_i;p_i) \geq \sum_i \ell(r_i;p_i)$. As s was arbitrary, r minimizes the separable objective.

Necessity and sufficiency together give $r \in R_S(p)$ if and only if the displayed inequalities hold at p, so $P_S(r)$ is exactly the set of beliefs satisfying them. $\square$

A.2 Squared-Distance Scoring

Proof of Proposition 1. Maximizing S_Q is equivalent to minimizing $\mathbb{E}_p\|r - \omega\|_2^2$. For fixed r,

$$\mathbb{E}_p\|r - \omega\|_2^2 = \|r - \mathbb{E}_p[\omega]\|_2^2 + \sum_{i=1}^{k} \mathrm{Var}_p(\omega_i).$$

The variance term is independent of r, and $\mathbb{E}_p[\omega] = np$. Hence optimal reports are exactly the feasible integer projections of np.

The full inverse projection condition is

$$\|r - np\|_2^2 \leq \|s - np\|_2^2 \quad \forall s \in \Omega,$$

or equivalently

$$2np \cdot (s - r) \leq \|s\|_2^2 - \|r\|_2^2.$$

Taking $s = r + e_i - e_j$, feasible when $r_j > 0$, gives

$$n(p_i - p_j) \leq r_i - r_j + 1.$$

Conversely, suppose all these single-count inequalities hold. For any $s \in \Omega$, write $d = s - r$. Pair each positive unit of d_i with a negative unit of d_j. Each coordinate j with $d_j < 0$ has $s_j < r_j$ and hence $r_j > 0$, so the single-count inequality $n(p_i - p_j) \leq r_i - r_j + 1$ is available for every paired transfer. Summing the single-count inequalities over the paired transfers gives

$$np \cdot d \leq r \cdot d + M,$$

where $M = \sum_{i:d_i>0} d_i$ is the number of transferred counts. Because d has integer components, $|d_i| \geq 1$ on its support, so

$$\sum_i d_i^2 \geq \sum_i |d_i| = 2M.$$

Doubling the displayed inequality and substituting $2M \leq \sum_i d_i^2 = \|d\|_2^2$ gives $2np \cdot d \leq 2r \cdot d + \|d\|_2^2$. Since $d = s - r$,

$$2np \cdot (s-r) \leq 2r \cdot (s-r) + \|s-r\|_2^2 = \|s\|_2^2 - \|r\|_2^2,$$

which is the full projection condition.

It remains to derive the coordinate bounds. Let $m = m(r) = |\{j : r_j > 0\}|$. If $r_i > 0$, summing

$$n(p_u - p_i) \leq r_u - r_i + 1$$

over all $u \neq i$ gives

$$1 - p_i \leq (k-1)p_i + \frac{n - kr_i + k - 1}{n},$$

so

$$p_i \geq \frac{r_i - 1}{n} + \frac{1}{nk}.$$

For the upper bound, sum

$$n(p_i - p_v) \leq r_i - r_v + 1$$

over positive-report coordinates $v \neq i$, and use nonnegativity of the zero-report coordinates:

$$1 - p_i \geq (m-1)p_i + \frac{n - mr_i - m + 1}{n}.$$

This gives

$$p_i \leq \frac{r_i + 1}{n} - \frac{1}{nm}.$$

If $r_i = 0$, the lower bound is 0. For the upper bound, sum

$$n(p_i - p_v) \leq 1 - r_v$$

over all m positive-report coordinates v:

$$1 - p_i \geq mp_i + \frac{n - m}{n},$$

so

$$p_i \leq \frac{m}{n(m+1)}.$$

The bounds are sharp, with an explicit attaining belief for each endpoint. For the lower bound with $r_i > 0$, take $p_i = (r_i - 1)/n + 1/(nk)$ and $p_u = r_u/n + 1/(nk)$ for $u \neq i$. This is a probability vector—it sums to one and every coordinate is at least $1/(nk)$—and $n(p_a - p_b) = r_a - r_b \leq r_a - r_b + 1$ for $a, b \neq i$ while $n(p_a - p_i) = r_a - r_i + 1$, so it lies in $P_Q(r)$ and the constraints with $b = i$ bind, attaining the lower endpoint. For the upper bound with $r_i > 0$, take $p_i = (r_i + 1)/n - 1/(nm)$, take $p_v = r_v/n - 1/(nm)$ for the positive-report coordinates $v \neq i$, and set $p_v = 0$ for the zero-report coordinates. This sums to one and is nonnegative, lies in $P_Q(r)$, and makes the constraints $n(p_i - p_v) \leq r_i - r_v + 1$ bind for every positive-report $v \neq i$, attaining the upper endpoint. For the upper bound with $r_i = 0$, take $p_i = m/(n(m+1))$, take $p_v = \{(m+1)r_v - 1\}/(n(m+1))$ for the positive-report coordinates v, and set $p_w = 0$ for the other zero-report coordinates. This sums to one and is nonnegative, lies in $P_Q(r)$, and makes the constraints $n(p_i - p_v) \le 1 - r_v$ bind, attaining the upper endpoint. Thus the displayed bounds are sharp. $\square$

A.3 Frequency-Guessing Scoring

Proof of Proposition 2. For any report r,

$$\mathbb{E}_p[S_0(r,\omega)] = a - b \Pr_p(\omega \neq r) = (a-b) + b \Pr_p(\omega = r),$$

so, since $b > 0$, expected-score maximization is equivalent to maximizing the multinomial probability mass $\Pr_p(\omega = r)$.

It remains to characterize the mode region. Compare r with $r + e_i - e_j$, which is feasible when $r_j > 0$. If $p_j = 0$ and $r_j > 0$, then r cannot be a mode unless all probability is zero, which is impossible. For $p_j > 0$,

$$\frac{\Pr_p(r + e_i - e_j)}{\Pr_p(r)} = \frac{r_j}{r_i + 1}\frac{p_i}{p_j}.$$

Thus a necessary condition for r to be a mode is

$$r_j p_i \le (r_i + 1)p_j \quad \forall i, j \text{ with } r_j > 0.$$

Conversely, if these inequalities hold, any feasible s can be reached from r by a sequence of single-count transfers from coordinates where the current vector exceeds s to coordinates where it is below s. At each step the same ratio is at most one, because source counts weakly decrease and destination counts weakly increase along the path. Therefore no feasible s has larger multinomial mass than r.

The coordinate upper bound follows by summing

$$r_j p_i \le (r_i + 1)p_j$$

over $j \neq i$:

$$(n - r_i)p_i \le (r_i + 1)(1 - p_i),$$

so $p_i \le (r_i + 1)/(n + 1)$. The lower bound follows by summing

$$r_i p_j \le (r_j + 1)p_i$$

over $j \neq i$:

$$r_i(1 - p_i) \le (n - r_i + k - 1)p_i,$$

so $p_i \ge r_i/(n + k - 1)$.

The bounds are sharp. The upper endpoint is attained by

$$p_i = \frac{r_i + 1}{n + 1}, \quad p_j = \frac{r_j}{n + 1} \quad (j \neq i).$$

The lower endpoint, when $r_i > 0$, is attained by

$$p_i = \frac{r_i}{n+k-1}, \quad p_j = \frac{r_j+1}{n+k-1} \quad (j \neq i).$$

When $r_i = 0$, the lower endpoint 0 is attained by setting $p_i = 0$ and assigning probabilities proportional to r_j on the reported support. When $r_i = n$, the upper endpoint 1 is attained by $p_i = 1$. $\square$

A.4 Comparing the Two Closed-Form Rules

Proof of Corollary 1. Throughout, $m = m(r)$ is the number of categories with $r_i > 0$, and the average coordinate width is $(1/k)\sum_{i=1}^{k}(\overline{p}_i - \underline{p}_i)$.

Part (1). By Proposition 2 the frequency-guessing coordinate width is

$$\frac{r_i+1}{n+1} - \frac{r_i}{n+k-1} = \frac{r_i(k-2)+(n+k-1)}{(n+1)(n+k-1)}.$$

Summing over i and using $\sum_i r_i = n$,

$$\sum_{i=1}^{k}\left(\overline{p}_{0,i} - \underline{p}_{0,i}\right) = \frac{(k-2)n + k(n+k-1)}{(n+1)(n+k-1)} = \frac{(k-1)(2n+k)}{(n+1)(n+k-1)};$$

dividing by k gives $W_0(n,k)$, which does not depend on r.

Part (2). By Proposition 1 a positive-report category has coordinate width $2/n - 1/(nm) - 1/(nk)$ and a zero-report category has width $m/[n(m+1)]$. With m positive and $k-m$ zero categories,

$$\sum_{i=1}^{k}\left(\overline{p}_{Q,i} - \underline{p}_{Q,i}\right) = m\left(\frac{2}{n} - \frac{1}{nm} - \frac{1}{nk}\right) + (k-m)\frac{m}{n(m+1)} = \frac{1}{n}\left[2m - 1 - \frac{m}{k} + \frac{m(k-m)}{m+1}\right];$$

dividing by k gives $W_Q(m;n,k)$, a function of m alone.

Part (3). Evaluating part (2) at $m = k$ gives $W_Q(k;n,k) = 2(k-1)/(nk)$, and

$$W_Q(k;n,k) > W_0(n,k) \iff 2(n+1)(n+k-1) > n(2n+k) \iff k(n+2) > 2,$$

which holds for all $n \geq 1$ and $k \geq 2$. Evaluating at $m = 1$ gives $W_Q(1;n,k) = (k-1)(k+2)/(2nk^2)$, and

$$W_Q(1;n,k) < W_0(n,k) \iff (k+2)(n+1)(n+k-1) < 2nk(2n+k).$$

The right side minus the left equals $n^2(3k-2) + nk(k-2) - (k+2)(k-1)$, whose coefficients in n are nonnegative for $k \geq 2$, so it is increasing in n; at $n = 2$ it equals $k^2 + 7k - 6 > 0$. Hence $W_Q(1;n,k) < W_0(n,k)$ for all $n \geq 2$ and $k \geq 2$. $\square$

A.5 Manhattan-Distance Scoring

Proof of Proposition 3. Linearity of expectation gives

$$\mathbb{E}_p\left[\sum_{i=1}^{k}|r_i-\omega_i|\right]=\sum_{i=1}^{k}\mathbb{E}_p|r_i-\omega_i|.$$

The i-th term depends only on the binomial marginal $W_i \sim \mathrm{Bin}(n, p_i)$. For

$$\ell_i(t;p_i)=\mathbb{E}|t-W_i|,$$

the marginal cost of increasing the reported count from t to $t+1$ is

$$\ell_i(t+1;p_i)-\ell_i(t;p_i)=2F_i(t;n,p_i)-1.$$

Moving one reported count from j to i changes expected Manhattan loss by

$$\{2F_i(r_i;n,p_i)-1\}-\{2F_j(r_j-1;n,p_j)-1\}.$$

Hence no feasible single-count transfer improves the report if and only if

$$F_i(r_i;n,p_i)\geq F_j(r_j-1;n,p_j)\quad \forall i,j \text{ with } r_i<n,\ r_j>0.$$

The increments $2F_i(t;n,p_i)-1$ are weakly increasing in t, so each coordinate loss $\ell_i(\cdot;p_i)$ is discrete-convex; Lemma 1 then applies, and the no-transfer condition just derived is sufficient as well as necessary—so $P_M(r)$ is the exact identified set.

The threshold representation follows by taking c between

$$\max_{j:r_j>0}F_j(r_j-1;n,p_j)\quad \text{and}\quad \min_{i:r_i<n}F_i(r_i;n,p_i).$$

Inverting the resulting binomial-CDF inequalities coordinate by coordinate gives the threshold-indexed box-simplex description used for Manhattan bounds. □

B Design Comparison Details

The design comparison is generated by the script `scripts/design_efficiency.py`. The final grid uses $n \in \{5, 10, 20, 50\}$, $k \in \{2, 3, 5, 10\}$, $\alpha \in \{0.1, 0.3, 1, 3, 10\}$, and 5,000 draws from the symmetric Dirichlet distribution for each cell. For each draw, and for each of the three rules in the comparison—frequency-guessing, squared-distance, and Manhattan—the script computes an optimal report and then bounds for the identified set generated by that report. Hamming and Chebyshev distance are excluded; Section 5 explains why their identified sets do not yield tractable sharp bounds.

Frequency-guessing coordinate bounds are closed form and mean bounds are linear programs over the transfer region. Squared-distance coordinate bounds are closed form and mean bounds are linear programs over the single-count transfer polytope. Manhattan coordinate bounds are sharp: the script solves the one-dimensional threshold optimization implied by Appendix A.5 with a monotone root-find. Manhattan mean bounds use the same threshold representation but are computed over a threshold grid with tolerance 10^{-4}.

The generated files are CSV tables under `outputs/design_exercise/`. These outputs are diagnostics for experimental design. They are not simulations of subject behavior and do not replace the mathematical characterization of $R_S(p)$ and $P_S(r)$.

Table 3 substantiates the claim in Section 4 that the number of categories k and the number of trials n are second-order. It reports win shares marginally by k and by n, averaging over the other two design dimensions. Against the variation across belief concentration α in Table 1—where the leading rule's win share runs from near zero to one—the dependence on k and n is slight.

	Average-coordinate			Linear-mean		
	Guessing	Squared	Manhattan	Guessing	Squared	Manhattan
By number of categories k						
2	0.72	0.25	0.03	0.72	0.25	0.03
3	0.59	0.37	0.04	0.34	0.51	0.15
5	0.58	0.35	0.07	0.49	0.42	0.09
10	0.59	0.31	0.10	0.56	0.33	0.11
By number of trials n						
5	0.60	0.34	0.06	0.57	0.33	0.10
10	0.61	0.32	0.07	0.57	0.33	0.11
20	0.62	0.31	0.06	0.53	0.38	0.09
50	0.65	0.30	0.05	0.44	0.47	0.09

Table 3: Win share marginally by number of categories k and number of trials n, averaging over the other two design dimensions. Relative to belief concentration (Table 1), the dependence is second-order: the coordinate target is nearly flat in both, and the only visible movements are the binary case $k = 2$ and a mild drift toward squared-distance scoring at large n for the mean target.

Robustness. Two implementation choices were checked and leave the comparison unchanged. First, the linear-functional results are robust to the outcome vector. The same run also evaluates the extreme-category outcome vector $x = (0, \ldots, 0, 1)$ in place of the evenly spaced $x_i = (i - 1)/(k - 1)$; the ranking is unchanged, with win shares 0.56, 0.34, and 0.10 for frequency-guessing, squared-distance, and Manhattan scoring, against 0.53, 0.38, and 0.10 for the evenly spaced vector. Second, optimal-report ties are a measure-zero event under the continuous Dirichlet draws: the final run recorded a tie rate of 0.0000 for all three rules, so the lexicographic convention used to select a report when $R_S(p)$ is not a singleton has no effect on the reported numbers.